

Matching a Physical Drum Model to a Recording

A perceptual distance in readable units, and what it establishes about a modal tom

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Abstract. Fitting a physical instrument model to a recording needs an objective that a person can read. This paper describes one: a recording and a render are reduced to the same set of perceptually named quantities — partial frequencies and per-partial ring times, a time-varying fractional-octave envelope, an amplitude envelope, pitch glide, attack balance, and coverage in both directions — and each term of the distance is reported in its own unit, so that a result can be read against a tolerance rather than only against another run. Adoption is decided per term. The reference reduction measures inter-channel delay and aligns before averaging, which on this recording is the difference between five detected partials and sixteen. The search is deterministic given its seed, checkpointed, and refuses to resume across a change in the model or the measurement.

Applied to a double-headed tom, the apparatus takes the distance from 32.6 at the shipped default to 11.6, bringing five of the nine terms below their thresholds — amplitude envelope, glide, attack balance and both coverage shares — and it localises what remains. What remains is a **tilt** in the decay structure: the model’s mid-band partials ring two to five times longer than the reference’s while its fundamental rings less than half as long, and the head-damping parameter, which scales all loss rates together, is fitted against the lower bound of its range. That is a specific, testable statement about the model’s damping range, and it is the kind of statement the apparatus exists to produce.

Working-paper status. The fitted numbers here come from one complete report, produced by a run that was stopped by hand at 47% of its restart budget so the machine could be given the next experiment. The figures and every number below are read from that report.

1. THE PROBLEM

Fitting a physically parameterised synthesiser to a recording is an optimisation whose objective is the whole difficulty. A waveform difference is unusable against a recording of a different physical instrument. A spectral distance is usable but opaque: it produces one number, and when that number stops improving there is nothing to say about what is still wrong.

The apparatus described here takes the opposite approach. Every term is a quantity a listener would name, in the unit that quantity is perceived in, and the sum exists only to order candidates — decisions are made term by term. The result is that a fit does not merely converge; it says what it has achieved and what it has not, and the latter turns into statements about the model.

2. THE INSTRUMENT AND THE TARGET

The instrument is a double-headed tom rendered by modal synthesis: two membranes coupled through a lumped air cavity, with a Berger tension nonlinearity, a microphone model, and a three-band stochastic attack layer covering the transient region modal synthesis reaches poorly. Its configuration is SI-valued and versioned; the product exposes seventeen normalised parameters over it. Excitation is selectable between a prescribed half-sine and a two-way coupled Hertzian contact with Hunt–Crossley damping [1], [2], whose contact time follows the strike velocity rather than being imposed [3], [4].

The target is one recorded tom hit. Its provenance is a sample library, with no documented microphone, room or processing chain, and no licence permitting redistribution. It is therefore a **timbre-match target**, not an acoustic validation reference: no automated test depends on it, and it is not part of the repository. Keeping that distinction sharp matters, because some of what the reduction reveals is a property of the recording rather than of any drum.

3. REDUCTION: FROM A SIGNAL TO NAMED QUANTITIES

Reference and candidate pass through the same extraction, so any asymmetry between them is a property of the signals rather than of the measurement.

Partials come from prominence-based peak picking on a 64k transform of the sustain window. Prominence rather than a bare local-maximum test, because ripples on the fundamental’s skirt satisfy the latter and are not modes. Each surviving peak carries a level, a ring time from a least-squares fit to its log envelope, and that fit’s R^2 — which later lets a partial’s frequency count while its decay does not, since a beating pair or a mode buried under its neighbour has a slope but not a meaningful one.

Spectral envelope is a 24-band third-octave vector computed over each of four windows spanning the hit — attack, early, body, tail — so it describes not only the timbre but the way the timbre evolves.

Amplitude envelope, pitch glide and **attack balance**, the click-to-body ratio, complete the reduction.

Deliberately absent is any sample-aligned waveform comparison. Against a recording of a different physical drum, waveform error measures the phase relationship between two signals that were never meant to share one: it is large for candidates that sound identical and small for candidates that do not. Its proper use, regression between two renders of the same model, is kept elsewhere.

4. THE DISTANCE

4.1. Units and weights

Each term is reported in the unit its own perception is measured in: partial frequency in cents, because the ear hears pitch as a ratio and 3 Hz means something different at 118 Hz than at 1180; partial decay as $|\ln(T_{60}^{\text{ref}}/T_{60}^{\text{cand}})|$, because ringing twice as long and half as long are the same size of mistake; levels and envelopes in dB; glide in cents.

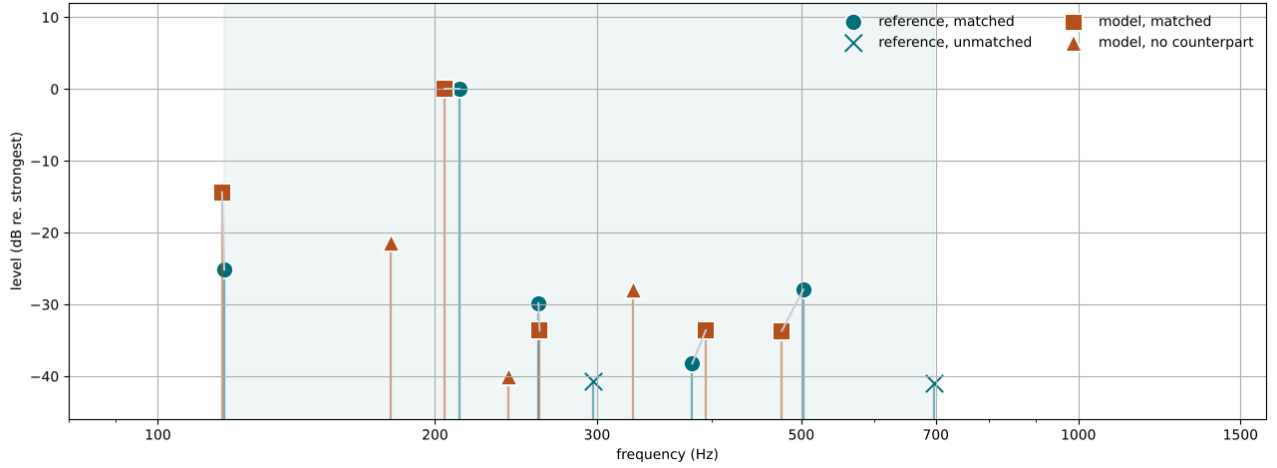


Figure 1: What the two coverage terms see. Grey links join matched pairs; shading marks the span between the lowest and highest reference partial, within which the spurious share is counted. The two reference partials without counterparts sit at -41 dB, close to the detection floor, and are correspondingly cheap to miss.

Weights are the reciprocal of each term’s “clearly wrong” threshold — 25 cents of pitch, 3 dB of partial balance, a factor of 1.4 in ring time, 4 dB of spectral shape, 3 dB of envelope, 40 cents of glide, 6 dB of attack balance — so a just-audible error anywhere contributes about equally to the sum. Both signals are peak-normalised, making every term gain-invariant; that is forced rather than chosen, since the reference is normalised and its loudness carries no information.

4.2. Matching

Partials are matched greedily by closeness in cents rather than in index order, so one badly placed candidate cannot cascade a misidentification through the series, and each candidate is claimed at most once, so it cannot explain two reference modes. The tolerance widens with mode index, since real two-headed drums scatter about $\pm 20\%$ around the ideal Bessel series in both directions [5].

4.3. Coverage, in both directions

Six terms are computed over matched pairs, and an error averaged over matched pairs is zero when there are no pairs. Each is therefore blended against a fixed penalty in proportion to the share of the reference left unmatched, so that a partial which is missing costs what a partial which is present but wrong costs.

That share must be weighted, and the weighting is the substantive choice. Weighting by **energy** concentrates it on whichever partial is loudest — on this reference, one partial carries 99.4% of the total — so a candidate holding that partial alone reports almost nothing missing. Weighting by **count** overcorrects: the reference contains a genuine, isolated component 41 dB down that no two-headed drum will produce, and failing to reproduce it must not cost what failing to reproduce the fundamental costs. Each partial is therefore worth **how far it stands above the detection floor, in dB** — zero at the floor, growing with prominence, monotone in loudness like energy but compressed enough that several missing quiet partials cannot be rounded away.

A second share scores the mirror case. Because matching iterates the reference, a candidate partial with no reference counterpart is invisible to every partial term and reaches the sum only through the spectral envelope, so a candidate can cover the reference completely while its second-loudest component is a mode the reference does not have. Both shares therefore appear, under the same weighting and at the same weight. Figure 1 is the picture they score.

The spurious share is counted **only** between the lowest and highest reference partial. Outside that span the reference’s own detection is unproven — a recording’s noise floor hides modes a model legitimately has — so a partial out there is charged by the spectral

envelope, on evidence, and not by coverage. Without that bound the term would fit the recording’s limitations rather than the drum.

4.4. The gate

The sum orders candidates; it does not decide them. Adoption is gated on three terms individually — partial frequency within 25 cents, partial decay within 0.25 in log ratio, spectral envelope within 4 dB — because a lower sum can be bought in terms nobody listens for.

That is not hypothetical. In one run the candidate with the **best** spectral envelope of any measured had the **fewest** partials, three, having found that raising the stochastic attack layer’s level and dropping its corner frequency into the band of interest satisfies a band-energy measure. Broadband noise can satisfy a spectral envelope; it cannot produce resolvable partials. On the total alone that run looked like the best match to the recording.

5. PREPARING THE REFERENCE

A stereo recording of one hit is two views of the same event, and they are generally not simultaneous. Summing them is then a comb filter: for a delay τ ,

$$H(f) = 2|\cos(\pi f \tau)| \quad (1)$$

with nulls at $(2k+1)/2\tau$ and maxima at k/τ — a fixed ± 6 dB spectral shape imposed by microphone geometry alone [6].

The reduction therefore measures the inter-channel lag by cross-correlation and aligns before averaging. It fires only when the pair is genuinely one signal twice — correlation at least 0.5, and clearly better than summing — and never on short or uncorrelated input. The measured lag and correlation are reported on the decoded reference, so the condition cannot pass unnoticed, and regression tests pin both directions: a delayed pair is aligned, an uncorrelated pair is left alone.

On this recording the second channel lags the first by 69 samples at 44.1 kHz, 1.56 ms, with correlation 0.942 at that lag against 0.361 at zero. Equation 1 then predicts nulls at 320, 959 and 1598 Hz and maxima at 639 and 1278 Hz, which is what the naive downmix shows against a single channel, with no fitted parameter (Figure 2).

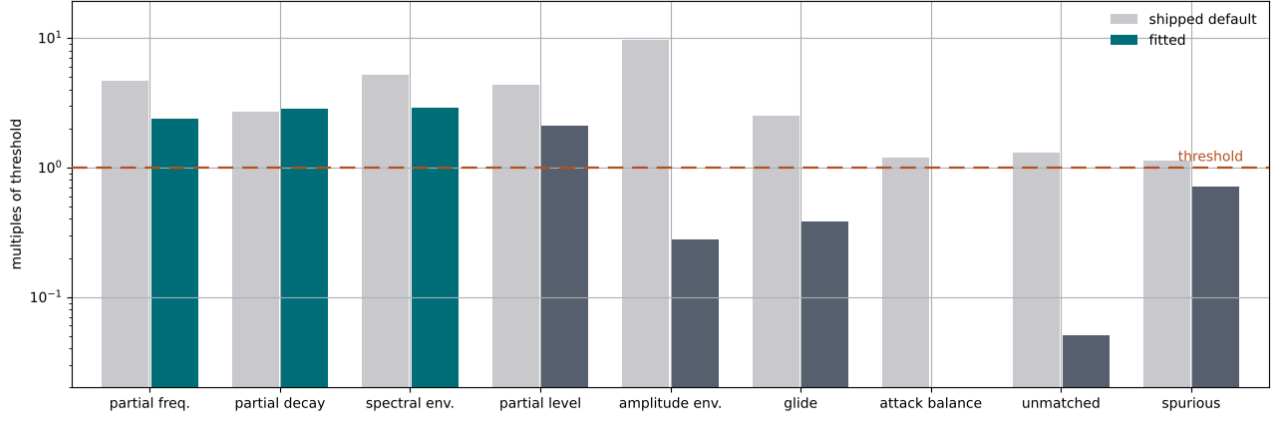


Figure 3: Every term as a multiple of its own threshold; the three gated terms are highlighted. Five terms are brought below threshold, three remain between $2.4\times$ and $2.9\times$, and the two coverage shares confirm the candidate is a drum rather than either degenerate extreme.

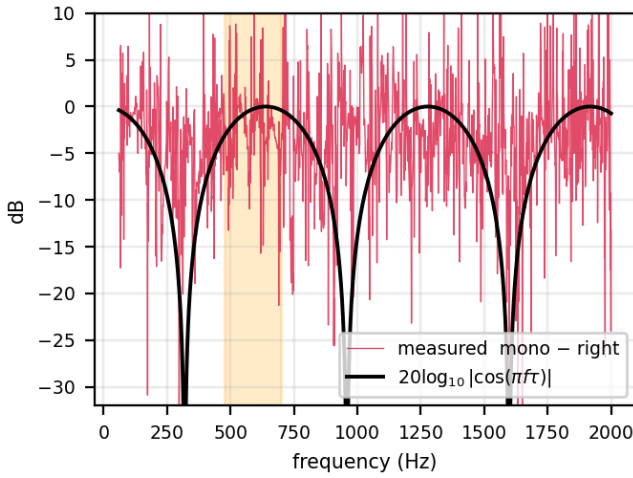


Figure 2: The naive mono downmix relative to the right channel alone, against Equation 1. Shaded: 476–700 Hz, where the comb’s first maximum falls.

The effect on the reduction is large, because the partial detector keeps a bounded number of peaks and the comb decides which survive:

Reference reduced as	Partials	In 476–700 Hz
mono, averaged naively	16	9
right channel alone	7	2
mono, aligned then averaged	5	2

Table 1: One recording, three reductions. Aligning before summing brings the mono view into agreement with the single channels; the naive sum agrees with neither.

Aligning also brings the reference’s ring times into agreement with membrane physics. Constant Q requires $T_{60} \propto 1/f$ [7]; the naive downmix gives a median of 1.13 s in the 476–700 Hz band where that law predicts 0.31 s, a factor of 3.6 that invites explanation in terms of a reverberant room. The aligned reduction gives 0.16, 0.19, 0.60 and 0.61 s, which needs no such explanation.

6. THE SEARCH

The distance is minimised with the Mayfly Optimization Algorithm [8], a build-time-only dependency imported by the fitting command and nothing else — the shipped binary is unchanged. The search space is exactly the bank the product exposes, plus strike velocity, with the quality tier pinned: mode count buys fidelity with CPU and is a product decision rather than a property of this drum.

Runs are deterministic given the seed. Progress is checkpointed with the completed restarts and the running best point, written atomically, and the checkpoint carries a fingerprint of the baseline cost. A resume across any change to the model or the measurement is refused rather than silently mixed — which has repeatedly caught a stale checkpoint that would otherwise have produced a run whose early generations were scored by one metric and its later ones by another.

One property of the model is worth stating here, because it determines the tier a comparison must run at. Mode selection retains a fixed **count** — 48, 96 or 160 by quality tier — while membrane mode frequencies scale as $\sqrt{T}$. Rendered bandwidth is therefore a function of tier **and tuning** together, and it falls as the drum is tuned down. For a bank at 331 N/m, roughly a quarter of the shipped batter tension:

Tier	Modes	Top mode	In 476–700 Hz
Draft	48	467 Hz	0
Standard — <i>shipped default</i>	96	664 Hz	45
High	160	852 Hz	58

Table 2: Modal ceiling of a low-tuned bank, by quality tier.

Two consequences follow: an instrument truncated by count loses its top octave when it is tuned down, and any comparison between configurations must run at the tier that ships. Bounding mode selection by frequency instead would make the ceiling independent of tuning.

7. WHAT THE FIT ESTABLISHES

The run reported here fits against the right channel at the shipped Standard tier with Hertzian contact: eight restarts of 150 iterations over a population of 16, seed 1. It was stopped by hand after 59,056 evaluations, 47% of that restart budget, so the machine could be turned to the next experiment; the report it wrote is complete, and everything below is read from it.

The distance falls from 32.585 at the shipped default to 11.630. More usefully, Figure 3 shows where it fell.

Five of the nine terms are brought below threshold: amplitude envelope to $0.28 \times$ (0.84 dB against 3), glide $0.38 \times$ (15.4 cents against 40), attack balance $0.008 \times$ (0.05 dB against 6), and both coverage shares — unmatched $0.05 \times$, spurious $0.71 \times$. The coverage pair is the useful confirmation: the candidate accounts for essentially all of the reference’s audible partial content while inventing little, so the remaining terms are being computed over a genuine correspondence rather than over one lucky pair. The reference reduces

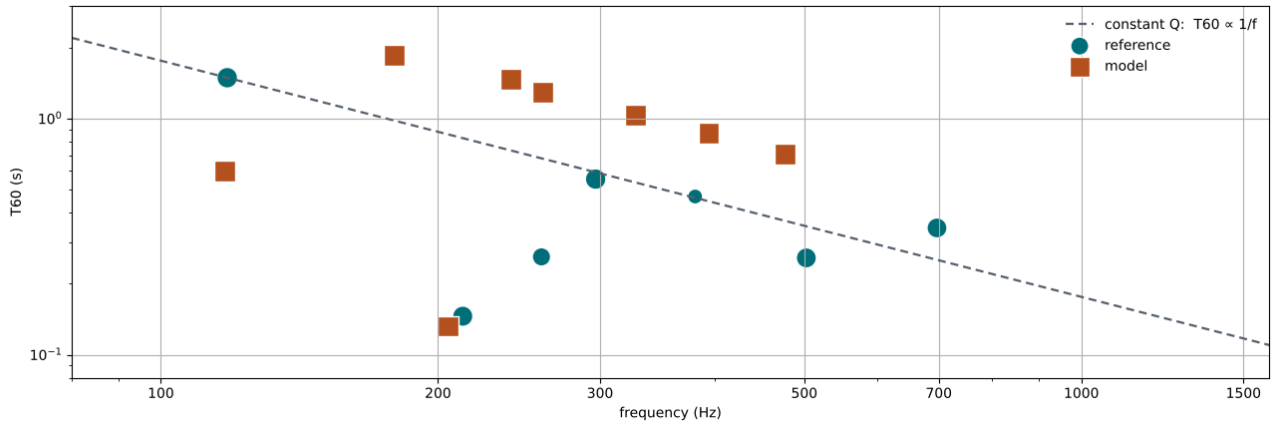


Figure 4: Ring time against frequency, with the constant- Q law anchored on the reference fundamental. The mismatch is a tilt rather than an offset: the model rings two to five times too long between 250 and 500 Hz, and less than half as long as the reference at the fundamental.

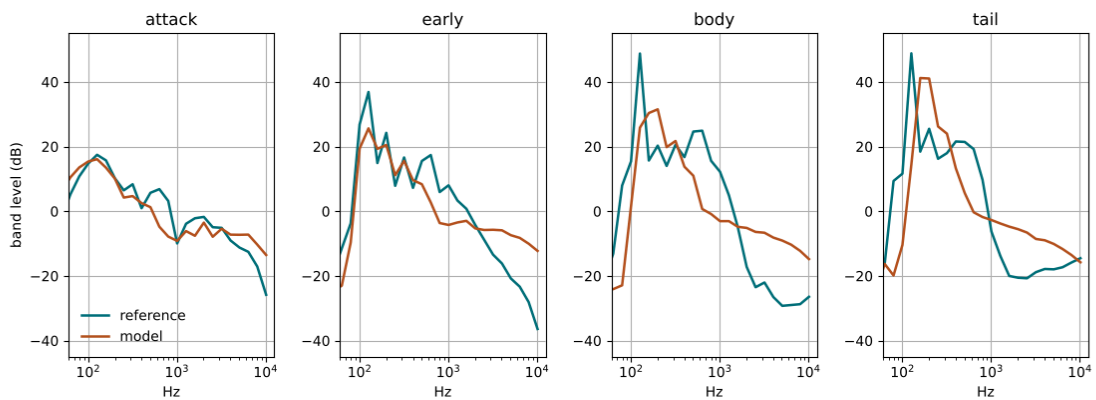


Figure 5: Spectral envelope, window by window. The attack matches within a few dB over most of the band; the discrepancy appears in the body and tail and is concentrated above 1.5 kHz, where the model runs 10–20 dB hot.

to seven partials and the candidate to eight; five pair up, the two reference partials left over sit at -41 dB, just above the detection floor.

Three gated terms remain above threshold — partial frequency $2.4 \times$ (59.7 cents against 25), partial decay $2.8 \times$ (0.99 in log ratio against 0.35, and against the tighter 0.25 its own gate applies), spectral envelope $2.9 \times$ (11.5 dB against 4) — and they are not three independent problems. Figure 4 and Figure 5 show one.

The remaining error is a decay tilt. Ring times do not sit uniformly above or below the reference's: between 250 and 500 Hz the candidate rings two to five times too long, while at the fundamental it rings 0.60 s against the reference's 1.49 s — too short. The same fact appears in the spectral envelope as a body and tail excess above 1.5 kHz, energy that should have decayed and has not, with the attack window matching within a few dB. Decay, not spectral shaping, is what the three gated terms are measuring.

The reference's own decay times are strikingly non-monotone in frequency: its loudest partial at 213 Hz dies in 0.146 s while its 118 Hz fundamental sustains 1.49 s. No single constant- Q loss law produces that ordering, which is why a fit that gets the mid band right must get the fundamental wrong, and the reverse. Locating the residual as a shape of the loss law rather than as a level is exactly the discrimination the per-term units were built for.

And the fit points at where the range runs out. Head damping is fitted to 0.276 on a range whose lower bound is 0.25 — normalised position 0.036, against the edge. Since that parameter scales all loss rates together, the search is asking for **less** overall loss than the bank can express, and it spent its remaining freedom elsewhere, tuning the batter head to 933 N/m and shifting the pickup. This is a concrete

and testable claim about the model rather than about the recording: **the head-damping range does not reach the loss structure this instrument exhibits.** Extending the range, and checking whether the partial-decay and spectral-envelope terms fall together, is the next experiment — already running — and the distance is arranged so that it will be a clear answer either way.

8. WHAT TO WATCH FOR

Four points generalise beyond this instrument, each learned by measurement.

Measure inter-channel lag before summing. A stereo pair of one event is one signal twice, and averaging imposes a fixed comb on every spectral quantity downstream. A low zero-lag correlation between two views of one close-miked hit is evidence that the channels are offset in time, not that there is a room on it.

Score coverage both ways, and weight the two sides equally. Either direction alone admits a degenerate optimum. Weighting the invented-partial side more heavily than the missing-partial side inverts the degeneracy rather than closing it — discarding the drum becomes the cheapest way to invent nothing, and a run under such a weighting duly converged on two partials. The pressure toward completeness and the pressure toward tidiness are one quantity seen from two sides; the asymmetry belongs in the blend, where it is already present, and not in the weights.

Read a pinned parameter as a statement about the range, not about the search. A value sitting on a bound is the most informative thing an optimiser produces: it says the best available answer lies outside the box. Report the normalised position alongside the

engineering value, so that the difference between “fitted to 0.276” and “fitted to the bound” is visible without opening the bank.

Gate per term, and confirm by ear. A distance can be monotone in quality — ranking candidates in the order a listener would — while being wrong about all of them in absolute terms. Ordering is much easier to get right than calibration, and a converged search reports the former while appearing to report the latter.

9. SCOPE AND REPRODUCIBILITY

The model, the feature extraction, the distance and the search are in `internal/physical`, `internal/physical/match` and `cmd/fit-physical`. Runs are deterministic given the seed and are excluded from continuous integration, since they take minutes and require a recording the repository does not contain. The channel-alignment behaviour, the audibility weighting and the relationship between the two coverage weights are pinned by tests that need no recording.

Figure 1, Figure 3, Figure 4 and Figure 5 are drawn by `tools/paper-figures` — just `paper-figures` — from a single fit report, which carries both feature sets in full, so they are reproducible from a committed artefact rather than from the recording, which is not redistributable. That command reproduces the distance’s own greedy partial matching rather than approximating it, so Figure 1 shows the pairing the score was actually computed over. Figure 2 is measured from the two channels directly and is not regenerable without them.

REFERENCES

- [1] K. H. Hunt and F. R. E. Crossley, “Coefficient of Restitution Interpreted as Damping in Vibroimpact,” *Journal of Applied Mechanics*, vol. 42, no. 2, pp. 440–445, 1975, doi: 10.1115/1.3423596.
- [2] F. Avanzini and D. Rocchesso, “Modeling Collision Sounds: Non-Linear Contact Force,” in *Proceedings of the COST-G6 Conference on Digital Audio Effects (DAFx-01)*, Limerick, Ireland, 2001, pp. 61–66.
- [3] S. Dahl, “Spectral Changes in the Tom-Tom Related to Striking Force,” *STL-QPSR*, vol. 38, no. 1, pp. 59–65, 1997.
- [4] A. Wagner, “Analysis of Drumbeats — Interaction Between Drummer, Drumstick and Instrument,” Master’s thesis, 2006.
- [5] P. Richardson, R. Toulson, and J. Nunn, “The Tuning of Two-Headed Drums,” *The Journal of the Acoustical Society of America*, vol. 131, no. 1, pp. 887–895, 2012.
- [6] J. Blauert, *Spatial Hearing: The Psychophysics of Human Sound Localization*, Revised. MIT Press, 1997.
- [7] N. H. Fletcher and T. D. Rossing, *The Physics of Musical Instruments*, 2nd ed. Springer, 1998. doi: 10.1007/978-0-387-21603-4.
- [8] K. Zervoudakis and S. Tsafarakis, “A Mayfly Optimization Algorithm,” *Computers & Industrial Engineering*, vol. 145, p. 106559, 2020, doi: 10.1016/j.cie.2020.106559.